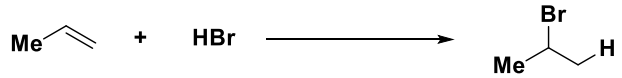
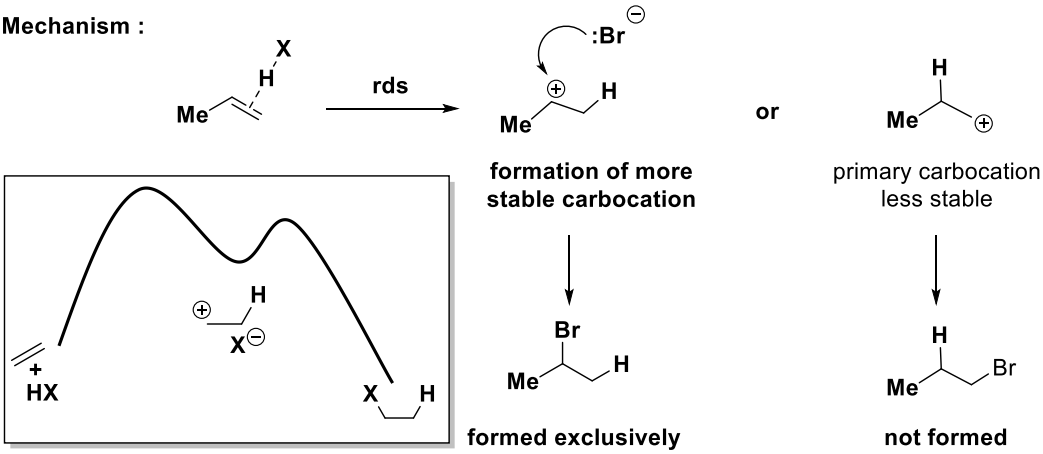
Class 3

- **Addition of HX :**

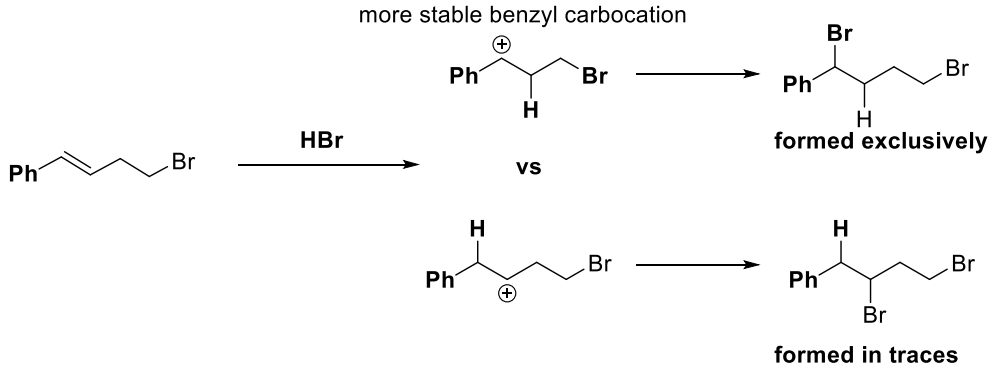


Mechanism :

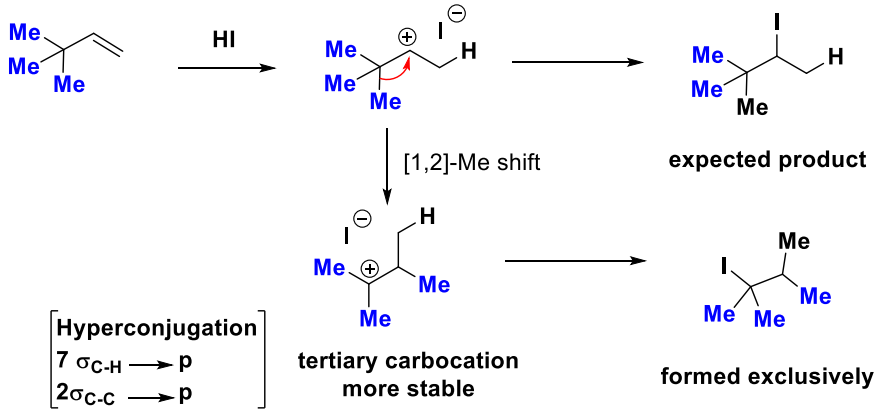


- Markonikov rule : The hydrogen add to the carbon of the double bond that has more hydrogen to start with or halogen or the electronegative moiety add to the more substituted carbon of the alkene.

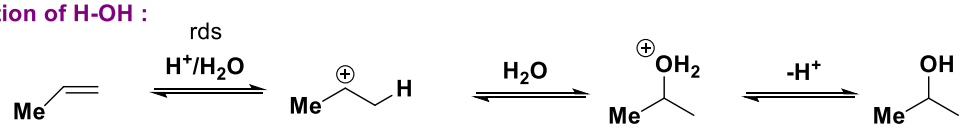
- **Understanding the reason :** <It is the stability of the carbocation that matters>



- **Carbocation can rearrange to give more stable carbocation** <Wagner–Meerwein rearrangement>

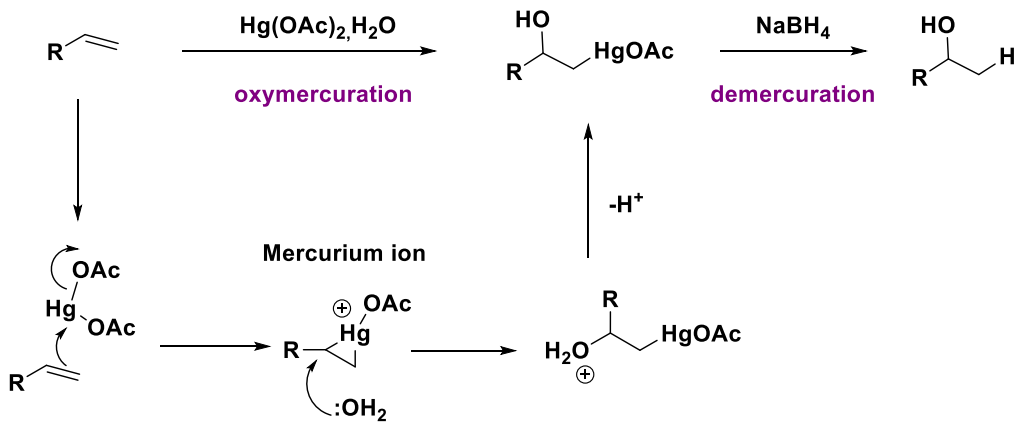


- **Addition of H-OH :**

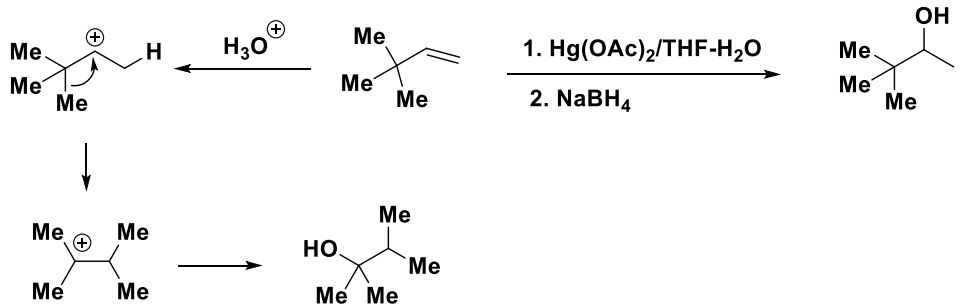


- formation of carbocation : directly or via π–complex formation is the slowest step, r.d.s.
- Markonikov addition
- Rearrangement of the carbocation possible
- Very difficult to predict wheather aq. acid will hydrate the alkene or dehydrate the alcohol.
- counter anion of the acid catalyst can add: preferred acids dil. H_2SO_4, H_3PO_4.....where the adducts can be hydrolyzed to give alcohol.

- **Oxymercuration-demercuration reaction :**

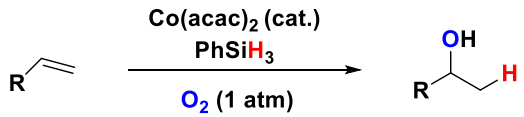


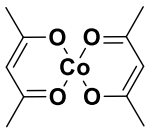
- Advantage : carbocation rearrangement does not happen.



Mercurie is highly toxic : not used in modern labs.

- **Mukaiyama Hydration : More practical**

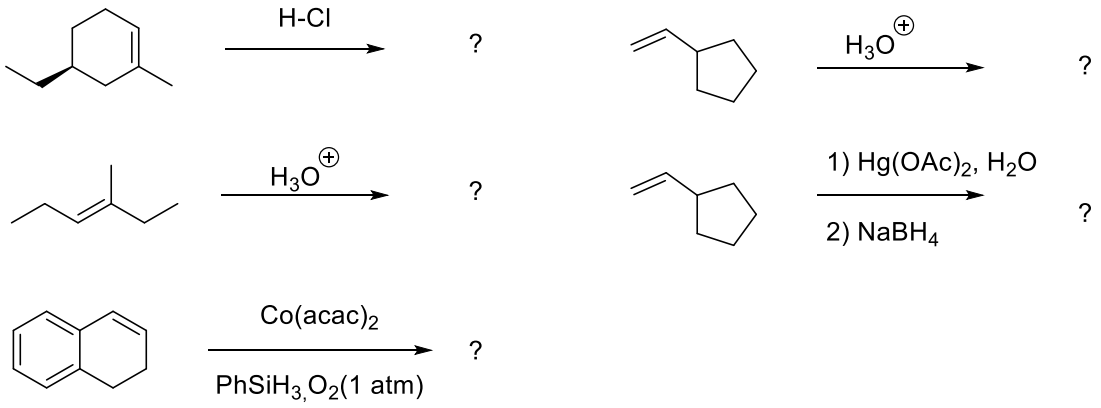
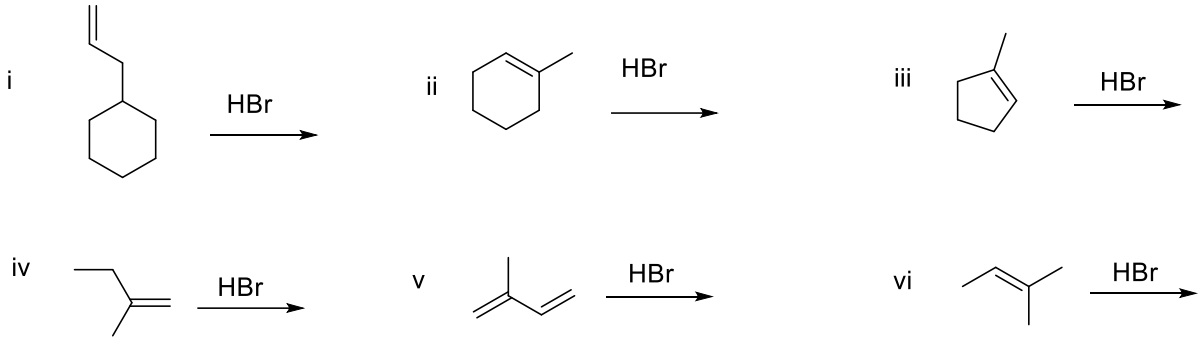




$Co(acac)_2$

Questions:

1. Show the intermediates and products that result from alkoxymercuration–demercuration of 1-methylcyclopentene, using methanol as the solvent.
2. When cyclohexene is treated with bromine in saturated aqueous sodium chloride, a mixture of ***trans*-2-bromocyclohexanol** and ***trans*-1-bromo-2-chlorocyclohexane** results. Propose a mechanism to account for these two products.
3. Draw the product(s) formed when CH3CH2CH2CH2CH=CH2 is treated separately with (a) H2O/H2SO4 and (b) BH3 then H2O2, KOH. Comment on the product(s) obtained in both the cases.
4. Predict the product with stereochemistry.



5. Write the product for the following HBr addition reaction.